Recovering Behavioral Axioms from Market Phenomena: An Indirect-Inference Framework for the Inverse Problem of Behavioral Finance

Xiangyu Chen
University of Chicago Booth School of Business
Xiangyu.Chen@chicagobooth.edu

March 2026

Abstract

We present a computational framework that solves the *inverse problem* of behavioral finance: given a set of observed market phenomena (stylized facts), recover the minimal set of behavioral assumptions that jointly generates them. The standard workflow in behavioral finance is *deductive*—one posits axioms and derives market implications. We invert this direction: from phenomena to axioms, with Occam's razor as the regularization criterion.

The framework couples an agent-based model built on Arrow–Debreu complete markets with an 11-dimensional theory-discriminating fingerprint vector and a scoring function grounded in the Simulated Method of Moments [SMM; Gourieroux et al., 1993, Duffie and Singleton, 1993]. The optimal weighting matrix $\mathbf{W}^* = \mathbf{\Sigma}^{-1}$, estimated from repeated baseline simulations with Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004], replaces the ad hoc distance metrics of prior agent-based approaches. A Pareto front over fit and model complexity, following Schmidt and Lipson [2009], provides a principled alternative to fixed parsimony penalties.

Five canonical behavioral theories—overconfidence [Daniel et al., 1998], loss aversion [Barberis et al., 2001], herding [Bikhchandani et al., 1992], noise trader risk [De Long et al., 1990], and the disposition effect [Shefrin and Statman, 1985]—are encoded as modular rules and evaluated against empirically calibrated targets. Phase 1 validation demonstrates correct theory recovery for all five rules, with Jacobian analysis revealing the sensitivity structure of the fingerprint space. Phase 2 joint testing evaluates all $2^5 = 32$ rule subsets to map the full interaction landscape: pairwise complementarity, redundancy, and contradiction among behavioral theories. The fingerprint incorporates the leverage effect of Cont [2001], and empirical calibration draws on Odean [1998], Frazzini [2006], and Sias [2004].

Contents

1 Introduction **3**

2 Related Work **4**
2.1 Behavioral Finance Theory 4
2.2 Agent-Based Computational Finance 4
2.3 Stylized Facts as Targets 4

2.4 Indirect Inference and Simulation-Based Estimation . . . 4
2.5 Parsimony and Automated Discovery . . . 5

3 Simulator **5**
3.1 Economy . . . 5
3.2 Preferences . . . 5
3.3 Optimal Demand . . . 6
3.4 Walrasian Equilibrium . . . 6
3.5 Per-Period Dynamics . . . 6
3.6 Composite Price and Volume . . . 7

4 Behavioral Rules **7**
4.1 Overconfidence . . . 8
4.2 Loss Aversion . . . 8
4.3 Herding . . . 8
4.4 Noise Trader Risk . . . 9
4.5 Disposition Effect . . . 9

5 Evaluator **9**
5.1 Theory Fingerprint . . . 9
5.2 Target Fingerprints . . . 11
5.3 Indirect Inference Scoring . . . 11
5.4 Pareto Front . . . 12
5.5 Jacobian Analysis . . . 12

6 Inductive Loop **13**

7 Validation Results **13**
7.1 Experimental Setup . . . 13
7.2 Baseline Fingerprint . . . 14
7.3 Theory Recovery . . . 14
7.4 Deviation from Baseline . . . 15
7.5 Jacobian Analysis . . . 16
7.6 Pareto Front . . . 17
7.7 Phase 2: Joint Testing . . . 17
7.8 J-Statistic and Overidentification . . . 19

8 Discussion **19**
8.1 What the Results Reveal . . . 19
8.2 The Role of the Weighting Matrix . . . 20
8.3 Design Choices and Trade-offs . . . 20
8.4 Limitations . . . 20
8.5 Future Directions . . . 21

9 Conclusion **21**

1 Introduction

The standard workflow in theoretical behavioral finance is *deductive*: one posits a set of behavioral axioms—loss aversion, overconfidence, herding—and derives their market-level implications. Decades of such work have produced a rich catalogue of behavioral theories, each explaining a subset of market anomalies. Yet the field faces a fundamental identification problem: multiple theories often predict the same stylized facts, and the interactions between jointly active behavioral mechanisms remain poorly understood.

This paper addresses the dual problem—*inductive* recovery of axioms from phenomena—formalized as follows.

> **Inverse Problem.** Given a target vector of market statistics $\mathbf{t} \in \mathbb{R}^K$ encoding stylized facts, find the smallest rule set $\mathcal{R}^*$ such that the simulated market fingerprint $\mathbf{c}(\mathcal{R}^*)$ is close to $\mathbf{t}$.

This is the behavioral-finance analogue of the programme articulated by Schmidt and Lipson [2009] for the natural sciences: instead of searching for symbolic equations that parsimoniously fit experimental data, we search for *behavioral mechanisms* that parsimoniously generate observed market phenomena. The philosophical direction is also the dual of Tao's AI-for-mathematics programme: instead of axioms ⇒ theorems (deduction), we pursue phenomena ⇒ axioms (induction). Occam's razor serves as the regularization criterion—fewer assumptions are always preferred.

Methodological foundation. Formally, the framework is an instance of *indirect inference* [Gourieroux et al., 1993]. The fingerprint vector $\mathbf{c}$ plays the role of the auxiliary statistic, and minimizing the SMM objective function over the space of behavioral rule sets is equivalent to searching for the data-generating behavioral mechanism. This reinterpretation connects the inductive loop to the statistical theory of simulation-based estimation [Duffie and Singleton, 1993] and motivates the use of the optimal weighting matrix $\mathbf{W}^* = \mathbf{\Sigma}^{-1}$, which assigns greater weight to fingerprint dimensions that are more precisely estimated under the baseline model.

The connection to indirect inference also clarifies the epistemological status of the exercise. Classical hypothesis testing asks: "Is theory A consistent with the data?" Indirect inference asks: "Which structural model, when simulated, produces auxiliary statistics closest to those observed?" We extend this to the behavioral domain: "Which behavioral assumptions, when embedded in an agent-based economy, produce a fingerprint closest to the target?" The shift from testing to recovery is what makes the problem *inverse*.

Contributions. The framework has three components:

1. A **simulator** that runs an agent-based Arrow–Debreu economy with pluggable behavioral rules (§3), building on the heterogeneous-agent tradition of Arthur et al. [1997] and Brock and Hommes [1998].
2. An **evaluator** that computes an 11-dimensional theory-discriminating fingerprint—incorporating the leverage effect of Cont [2001]—and scores it against target fingerprints using the SMM quadratic form (§5).
3. A **rule generator** that proposes candidate behavioral rules informed by evaluator feedback, with model complexity controlled via Pareto-front tracking rather than a fixed penalty (§6).

Section 2 situates the framework within the existing literature. Section 4 details the five base behavioral rules and their empirical calibration. Section 7 presents Phase 1 validation results, including Jacobian analysis and Pareto-front construction. Section 8 discusses limitations and future directions.

2 Related Work

The framework sits at the intersection of four research strands.

2.1 Behavioral Finance Theory

The five canonical behavioral theories we encode as rules each emerged from a distinct programme. Kahneman and Tversky [1979] introduced prospect theory, establishing loss aversion and probability weighting as departures from expected utility; Barberis et al. [2001] embedded these in an equilibrium asset-pricing model. Daniel et al. [1998] modelled overconfidence as excessive weight on private signals, generating short-horizon overreaction and long-horizon reversal. Bikhchandani et al. [1992] formalized informational cascades, with Sias [2004] documenting institutional herding empirically. De Long et al. [1990] showed that correlated noise-trader sentiment creates systematic risk that rational arbitrageurs cannot eliminate. Shefrin and Statman [1985] identified the disposition to sell winners and hold losers, with Odean [1998] and Frazzini [2006] providing extensive empirical evidence.

Each of these theories was developed and tested in isolation. Our framework asks: which subset is *necessary and sufficient* to generate a given set of stylized facts?

2.2 Agent-Based Computational Finance

The agent-based modelling tradition in finance begins with the Santa Fe Artificial Stock Market [Arthur et al., 1997], which demonstrated that heterogeneous adaptive agents can endogenously generate many features of real markets. Brock and Hommes [1998] introduced the discrete-choice switching mechanism, showing that even simple heterogeneity in expectations—fundamentalists versus chartists—can produce complex dynamics including "rational routes to chaos." Hommes [2006] and LeBaron [2006] survey the subsequent literature.

Our simulator departs from this tradition in one important respect: we use Arrow–Debreu securities with a Walrasian auctioneer rather than ad hoc price-impact functions or market-maker rules. This yields analytic demand functions and a well-defined equilibrium concept, at the cost of abstracting from microstructure.

2.3 Stylized Facts as Targets

Cont [2001] catalogued eleven "stylized facts" of financial returns—including fat tails, volatility clustering, the leverage effect, and absence of linear autocorrelation—that any plausible model should reproduce. These facts serve as the phenomena in our inverse problem. Our 11-dimensional fingerprint vector draws directly on Cont's catalogue, and the addition of the leverage effect ($\mathrm{Corr}(r_t, r_{t+1}^2) < 0$) as a discriminating dimension is motivated by his observation that this asymmetry is one of the most robust empirical regularities.

2.4 Indirect Inference and Simulation-Based Estimation

The Simulated Method of Moments [Duffie and Singleton, 1993] and indirect inference [Gourieroux et al., 1993] provide the econometric foundation for our scoring function. In classical SMM, one estimates structural parameters by minimizing the distance between auxiliary statistics computed from observed data and from simulated data, weighted by the inverse of the auxiliary statistics'

covariance matrix. We apply the same logic, but the "structural parameters" are not numerical values—they are *behavioral rules*. The search space is discrete (which rules to include) and combinatorial, rather than continuous.

The optimal weighting matrix $\mathbf{W}^* = \mathbf{\Sigma}^{-1}$ emerges naturally from the GMM theory of Hansen [1982]: it is the matrix that minimizes the asymptotic variance of the SMM estimator. We estimate $\mathbf{\Sigma}$ from repeated baseline simulations and regularize with Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004] to ensure positive definiteness in finite samples.

2.5 Parsimony and Automated Discovery

The Minimum Description Length (MDL) principle [Rissanen, 1978] provides the information-theoretic foundation for our Occam penalty: the best model is the one that achieves the shortest joint description of the model and the data it fails to explain. Schmidt and Lipson [2009] operationalized a similar principle for the natural sciences, using symbolic regression to discover compact mathematical laws from experimental data. Their key innovation was maintaining a *Pareto front* over fit and complexity, rather than committing to a fixed trade-off parameter. We adopt the same strategy: instead of a single penalty $\lambda \cdot n_{\text{assumptions}}$, we track the set of non-dominated (fit, complexity) pairs across all candidate rule sets.

3 Simulator

3.1 Economy

We consider a discrete-time Arrow–Debreu economy with $N = 100$ agents, $S = 3$ states of the world (Bad, Neutral, Good), and $T = 1{,}000$ trading periods. Each period, nature draws a state $s_t \in \{0, 1, 2\}$ according to the true probability vector $\boldsymbol{\pi}^* = (0.25, 0.50, 0.25)$. Agents trade a complete set of Arrow–Debreu securities that pay one unit of consumption in exactly one state and zero otherwise.

State-contingent endowments are

$$E = (E_0, E_1, E_2) = (850, 1000, 1150),$$

scaled so that $\mathbb{E}[E_s] = \sum_s \pi_s^* E_s = 1000$, which equals the initial wealth w_0 of every agent.

The Arrow–Debreu framework, following the foundational work of Arrow and Debreu [1954], provides a clean separation between preferences and market structure. Unlike order-book or market-maker models common in the ABM literature [Arthur et al., 1997, LeBaron, 2006], our setup yields analytic demand functions and a well-defined equilibrium concept, isolating behavioral distortions from microstructure effects.

3.2 Preferences

Agents have constant relative risk aversion (CRRA) utility:

$$u(c; \gamma) = \begin{cases} \dfrac{c^{1-\gamma}}{1-\gamma}, & \gamma \neq 1, \\ \ln c, & \gamma = 1, \end{cases}$$

where $\gamma_i \sim \mathcal{N}(4.0,\, 1.0)$ is drawn independently for each agent and clipped to $[0.5, \infty)$. Heterogeneous risk aversion is essential: it prevents the economy from collapsing to a representative-agent equilibrium where behavioral distortions are masked. This echoes the insight of Brock and Hommes [1998] that heterogeneity is a necessary condition for complex dynamics.

3.3 Optimal Demand

Given state prices $\mathbf{q} = (q_0, q_1, q_2)$ with $q_s > 0$ and $\sum_s q_s = 1$, agent i maximizes

$$\max_{c_{i,0},\, c_{i,1},\, c_{i,2}} \sum_{s=0}^{2} \pi_{i,s}\, u(c_{i,s};\, \gamma_i) \quad \text{s.t.} \quad \sum_{s=0}^{2} q_s\, c_{i,s} = w_i,$$

where $\pi_{i,s}$ is agent i's subjective probability of state s and w_i is wealth. The first-order conditions yield the analytic solution

$$c^*_{i,s}(\mathbf{q}) = \frac{w_i\, \pi_{i,s}^{1/\gamma_i}\, q_s^{-1/\gamma_i}}{Z_i}, \qquad Z_i = \sum_{k=0}^{2} \pi_{i,k}^{1/\gamma_i}\, q_k^{(\gamma_i - 1)/\gamma_i}. \tag{1}$$

The closed-form demand avoids per-agent numerical optimization, which would be prohibitive inside the equilibrium solver.

3.4 Walrasian Equilibrium

Equilibrium state prices satisfy the market-clearing condition:

$$\sum_{i=1}^{N} c^*_{i,s}(\mathbf{q}) = E_s, \qquad s = 0, 1, 2.$$

We parameterize prices via a free vector $\mathbf{x} \in \mathbb{R}^{S-1}$:

$$\mathbf{q} = \mathrm{softmax}\big([0,\ x_1,\ x_2]\big),$$

which automatically enforces $q_s > 0$ and $\sum_s q_s = 1$. The equilibrium problem becomes an unconstrained minimization:

$$\min_{\mathbf{x} \in \mathbb{R}^2} \sum_{s=0}^{2} \Big(\sum_{i=1}^{N} c^*_{i,s}\big(\mathrm{softmax}(\mathbf{x})\big) - E_s \Big)^2. \tag{2}$$

We solve (2) using L-BFGS-B with a robust multi-start strategy:

1. **Warm start**: initialize from the previous period's solution, exploiting temporal smoothness.
2. **Theory-informed cold start**: initialize at the representative-agent solution with $\gamma = \bar{\gamma}$.
3. **Zero cold start**: $\mathbf{x} = \mathbf{0}$ (uniform prices).

The solver accepts the first solution achieving function tolerance $\leq 10^{-15}$ and gradient tolerance $\leq 10^{-12}$.

3.5 Per-Period Dynamics

Algorithm 1 summarizes the per-period flow.

The order of operations—decay, Bayesian update, then behavioral modification—cleanly separates rational information processing from behavioral distortions, allowing each rule to operate on post-Bayesian beliefs.

Algorithm 1 Per-period simulation step

Input: agents $\{i\}$, market state, behavioral rules $\mathcal{R}$
for each agent i **do**
  Belief decay: $\boldsymbol{\pi}_i \leftarrow \frac{1}{2}\boldsymbol{\pi}_i + \frac{1}{2}\boldsymbol{\pi}^*$ ▷ revert toward true probs
end for
Draw true state $s_t \sim \boldsymbol{\pi}^*$
Generate signal: $\sigma_t = s_t$ with prob 0.6; else uniform draw from $\{0, 1, 2\} \setminus \{s_t\}$
for each agent i **do**
  Bayesian update: $\pi_{i,s} \propto \pi_{i,s} \cdot \ell(s \mid \sigma_t)$
  for each rule $r \in \mathcal{R}$ **do**
    Apply behavioral modification: $\boldsymbol{\pi}_i \leftarrow r(\boldsymbol{\pi}_i, \text{agent}_i, \text{market})$
  end for
  Clip and renormalize: $\boldsymbol{\pi}_i \leftarrow \text{clip}(\boldsymbol{\pi}_i, 10^{-6})/\|\cdot\|_1$
end for
Solve equilibrium (2) $\Rightarrow$ state prices $\mathbf{q}$
Compute optimal consumption via (1)
Settle: $w_i \leftarrow c^*_{i,s_t}$
Record composite price $P_t = \sum_s q_s E_s$, return, volume

3.6 Composite Price and Volume

The composite asset price is defined as

$$P_t = \sum_{s=0}^{2} q_{t,s}\, E_s,$$

representing the price of a claim to the aggregate endowment. Returns are computed as $r_t = (P_t - P_{t-1})/P_{t-1}$. Volume is measured as the total absolute rebalancing:

$$V_t = \frac{1}{2}\sum_{i=1}^{N} \left| c_i^{*,\text{new}} - c_i^{*,\text{old}} \right|.$$

The first 100 periods are discarded as warmup, leaving 900 periods for fingerprint computation.

4 Behavioral Rules

Each behavioral rule is a self-contained module implementing a common interface:

$$r : (\text{Agent}, \text{MarketState}) \longrightarrow \text{dict of attribute modifications.} \tag{3}$$

Rules are stateless—all information is passed via arguments—and composable: multiple rules can be applied sequentially within each period. This modularity is what makes the inverse problem tractable: the search space is the power set of available rules, and each element can be evaluated independently. Table 1 summarizes the five base rules.

Table 1: Summary of base behavioral rules.

Rule	Mechanism	n_p	Source Theory	Empirical Calibration
Overconfidence	Posterior temperature scaling	1	DHS (1998)	Ben-David & Hirshleifer (2012)
Loss aversion	Ref.-dependent belief distortion	1	BHS (2001)	Kahneman & Tversky (1979)
Herding	Beliefs chase empirical freq.	1	BHW (1992)	Sias (2004)
Noise trader	AR(1) sentiment process	2	DSSW (1990)	—
Disposition	Asymmetric gain/loss updates	1	SS (1985)	Odean (1998); Frazzini (2006)

4.1 Overconfidence

Following Daniel et al. [1998], overconfident agents exaggerate the precision of their private signals. We implement this as posterior temperature scaling:

$$\tilde{\pi}_{i,s} = \frac{\pi_{i,s}^{1/\tau}}{\sum_k \pi_{i,k}^{1/\tau}}, \qquad \tau = 0.55.$$

Since $\tau < 1$, the exponent $1/\tau \approx 1.82$ sharpens the posterior distribution, making agents act as if their information is more precise than it actually is. Ben-David and Hirshleifer [2012] document that individual investors exhibit precisely this pattern: they trade as if their private valuations are 10–30% more precise than warranted, consistent with $\tau \in [0.45, 0.6]$.

Expected signature: Overreaction to signals produces negative short-lag autocorrelation, fat tails (elevated excess kurtosis), negative skewness, and large drawdowns as prices overshoot fundamentals.

4.2 Loss Aversion

Following Barberis et al. [2001], who embed prospect theory [Kahneman and Tversky, 1979] in an equilibrium asset-pricing model, agents in the loss domain overweight the probability of bad outcomes:

$$\tilde{\pi}_{i,0} = \begin{cases} \pi_{i,0}\,(1 + \delta\,\lambda\,|g_i|), & g_i < 0 \text{ (loss domain)}, \\ \pi_{i,0}\,\max(1 - \delta \cdot 0.5 \cdot g_i,\ 0.5), & g_i \geq 0 \text{ (gain domain)}, \end{cases}$$

where $g_i = (w_i - w_i^{\text{ref}})/w_i^{\text{ref}}$ is the fractional gain/loss relative to the reference point, $\lambda = 2.25$ is the loss-aversion coefficient calibrated to Kahneman and Tversky [1979], and $\delta = 0.15$ is the distortion strength. The gain-domain branch implements a house-money effect: agents who have accumulated gains become slightly more risk-tolerant.

Expected signature: High GARCH persistence (time-varying risk premium), positive skewness (fewer crashes due to loss-induced caution), and long drawdown durations.

4.3 Herding

Following Bikhchandani et al. [1992], agents shift their beliefs toward the recent empirical state frequency, creating information cascades:

$$\tilde{\boldsymbol{\pi}}_i = (1 - H)\,\boldsymbol{\pi}_i + H\,\hat{\boldsymbol{\pi}}_{\text{emp}}, \qquad H = 0.47,$$

where $\hat{\boldsymbol{\pi}}_{\text{emp}}$ is the empirical frequency of realized states over the last $L = 5$ periods. Sias [2004] documents that institutional investors exhibit a herding correlation of approximately 0.49 across quarterly holdings changes, providing empirical support for the magnitude of H.

Expected signature: Pronounced lag-5 autocorrelation (cascade followed by reversal at the look-back horizon), high volume–return correlation from coordinated belief shifts, and elevated GARCH persistence.

4.4 Noise Trader Risk

Following De Long et al. [1990], all agents are subject to a common AR(1) sentiment process that distorts beliefs in logit space:

$$s_t = \rho\, s_{t-1} + \epsilon_t, \qquad \epsilon_t \sim \mathcal{N}(0, \sigma^2_{\text{noise}}), \qquad \rho = 0.9,\ \sigma_{\text{noise}} = 1.0,$$

$$\log \tilde{\pi}_{i,s} = \log \pi_{i,s} + s_{t,s},$$

with softmax renormalization. The sentiment is market-wide (seeded by period, not agent), so all agents share the same irrational belief perturbation each period. With $\rho = 0.9$ the half-life is approximately 6.6 periods, creating sustained mispricing: when all agents share the same persistent sentiment, there is no rational arbitrageur to offset the distortion, generating excess volatility and fat tails. Sentiment state is stored on each agent across periods, while belief decay (Section 3.5) resets only the base belief vector, preserving sentiment persistence.

Expected signature: High excess kurtosis (fat tails from persistent sentiment extremes), elevated vol–return correlation, reduced lag-1 autocorrelation (prices track sentiment rather than mean-reverting to fundamentals), and high GARCH persistence.

4.5 Disposition Effect

Following Shefrin and Statman [1985], agents in the gain domain suppress the good-state probability (eager to realize gains), while agents in the loss domain inflate it (reluctant to realize losses):

$$\tilde{\pi}_{i,s_{\max}} = \begin{cases} \pi_{i,s_{\max}} \cdot \max(1 - \alpha\, g_i,\ 0.3), & g_i > 0, \\ \pi_{i,s_{\max}} \cdot (1 + \alpha \cdot 0.5 \cdot |g_i|), & g_i \le 0, \end{cases}$$

where $s_{\max} = \arg\max_s E_s$ identifies the best state and $\alpha = 0.2$. Odean [1998] provides extensive evidence that individual investors are approximately 1.5 times more likely to sell winning stocks than losing ones. Frazzini [2006] shows that the disposition effect leads to predictable underreaction in stock prices, with a spread of approximately 243 basis points per month between stocks with unrealized capital gains and losses.

Expected signature: Negative autocorrelation, contrarian volume–return correlation (negative), negative skewness, and the longest drawdown durations among all rules.

5 Evaluator

5.1 Theory Fingerprint

The theory fingerprint is a vector $\mathbf{c} \in \mathbb{R}^{11}$ computed from the simulated time series $(r_t, V_t, P_t, \sigma_t, s_t)$. Each dimension is chosen to discriminate among the five target behavioral theories. The selection of dimensions is guided by the empirical regularities catalogued by Cont [2001], who identifies eleven stylized facts of financial returns that any plausible model should reproduce. Table 2 defines all components.

Table 2: Fingerprint dimensions and computation.

Dim	Name	Definition
1	`autocorr_lag1`	Sample autocorrelation of returns at lag 1
2	`autocorr_lag5`	Sample autocorrelation at lag 5
3	`autocorr_lag21`	Sample autocorrelation at lag 21
4	`vol_return_corr`	Pearson correlation between $\lvert r_t \rvert$ and V_t
5	`garch_alpha_beta`	$\hat{\alpha} + \hat{\beta}$ from GARCH(1,1) MLE
6	`return_skewness`	Fisher skewness of return distribution
7	`excess_kurtosis`	Fisher excess kurtosis of returns
8	`price_impact`	OLS slope: $r_t = \beta\, d_t + \varepsilon_t$
9	`max_drawdown`	Maximum peak-to-trough decline
10	`drawdown_duration`	Periods from peak until recovery
11	`leverage_effect`	$\mathrm{Corr}(r_t,\, r_{t+1}^2)$ [Cont, 2001]

Autocorrelation. The sample autocorrelation at lag k is

$$\hat{\rho}_k = \frac{\sum_{t=k}^{T-1}(r_t - \bar{r})(r_{t-k} - \bar{r})}{\sum_{t=0}^{T-1}(r_t - \bar{r})^2}.$$

Cont [2001] notes that linear autocorrelations of returns are typically insignificant except at very short (intraday) lags, while absolute returns exhibit significant positive autocorrelation over long lags—the volatility clustering phenomenon.

GARCH(1,1) estimation. We fit

$$\sigma_t^2 = \omega + \alpha\, r_{t-1}^2 + \beta\, \sigma_{t-1}^2$$

via maximum likelihood (Nelder–Mead), returning $\hat{\alpha} + \hat{\beta}$ as the volatility persistence measure. If optimization fails, we fall back to the first-order autocorrelation of squared returns.

Leverage effect. Following Cont [2001], we compute

$$\mathcal{L} = \mathrm{Corr}(r_t,\, r_{t+1}^2),$$

which measures the asymmetric dependence between returns and subsequent volatility. The empirical regularity is $\mathcal{L} < 0$: negative returns are associated with higher future volatility. This dimension was absent from our original 10-dimensional fingerprint; its addition improves discrimination between theories that produce symmetric versus asymmetric return-volatility dynamics.

Price impact. The signal direction $d_t \in \{-1, 0, +1\}$ maps Bad/Neutral/Good signals. The OLS coefficient $\hat{\beta}$ measures how strongly prices respond to public information.

Drawdown morphology. Maximum drawdown is $\mathrm{DD}_{\max} = \max_t (P_{\mathrm{peak},t} - P_t)/P_{\mathrm{peak},t}$. Duration counts the periods from peak until the price series sets a new high.

5.2 Target Fingerprints

Target fingerprints are derived from the theoretical predictions of each canonical behavioral model. Each target represents the direction and magnitude that a specific behavioral mechanism is expected to push the fingerprint away from the rational baseline. Table 3 presents the full target matrix.

Table 3: Target fingerprint vectors for five behavioral theories.

Dimension	Overconf.	Loss Av.	Herding	Noise	Disp.
autocorr_lag1	−0.28	−0.35	−0.27	−0.32	−0.33
autocorr_lag5	0.00	−0.01	−0.18	−0.02	−0.03
autocorr_lag21	0.01	0.03	0.01	0.04	0.07
vol_return_corr	−0.05	−0.05	+0.10	0.06	−0.08
garch_alpha_beta	0.86	0.97	0.91	0.58	0.98
return_skewness	−0.30	+0.15	−0.05	0.10	−0.30
excess_kurtosis	+0.20	−0.50	−0.50	−0.20	−0.60
price_impact	0.055	0.025	0.020	0.032	0.024
max_drawdown	0.30	0.09	0.13	0.12	0.11
drawdown_duration	620	850	530	560	1000
leverage_effect	−0.15	−0.20	+0.05	−0.02	−0.10

5.3 Indirect Inference Scoring

The connection between our fingerprint-based scoring and indirect inference [Gourieroux et al., 1993] is direct. In SMM, the objective function is the quadratic form

$$Q(\mathcal{R}) = \big[\mathbf{c}(\mathcal{R}) - \mathbf{t}\big]^\top \mathbf{W} \big[\mathbf{c}(\mathcal{R}) - \mathbf{t}\big], \tag{4}$$

where $\mathbf{c}(\mathcal{R})$ is the fingerprint computed from a simulation with rule set $\mathcal{R}$, $\mathbf{t}$ is the target, and $\mathbf{W}$ is a positive-definite weighting matrix.

Optimal weighting matrix. Following Duffie and Singleton [1993], the optimal weighting matrix is the inverse of the long-run covariance matrix of the auxiliary statistics:

$$\mathbf{W}^* = \mathbf{\Sigma}^{-1}, \qquad \mathbf{\Sigma} = \mathrm{Var}\big[\mathbf{c}(\emptyset)\big], \tag{5}$$

where $\mathbf{c}(\emptyset)$ denotes the fingerprint under the baseline (no-rule) model. This matrix assigns greater weight to dimensions that are more precisely estimated under the null, and lower weight to noisy dimensions whose variation is dominated by simulation randomness.

We estimate $\mathbf{\Sigma}$ from $N_b = 20$ baseline simulations with distinct random seeds. Because $K = 11$ dimensions and $N_b = 20$ yields a potentially ill-conditioned sample covariance, we apply Ledoit–Wolf shrinkage [Ledoit and Wolf, 2004]:

$$\hat{\mathbf{\Sigma}}_{\mathrm{reg}} = (1-\alpha)\,\hat{\mathbf{\Sigma}} + \alpha\,\mathrm{diag}(\hat{\mathbf{\Sigma}}), \qquad \alpha = \min\big(0.2,\ 2/N_b\big), \tag{6}$$

which shrinks the sample covariance toward its diagonal, ensuring positive definiteness.

Fit score. The raw SMM distance $Q(\mathcal{R})$ is converted to a fit score in $[0, 1]$:

$$\text{FitScore} = \max\left(1 - \frac{\sqrt{Q/K}}{3},\ 0\right), \tag{7}$$

where the normalization by K and the scaling factor of 3 are chosen so that a "typical" deviation from target maps to a fit score near zero, while a near-perfect match yields a score near one.

Occam penalty. The final score penalizes model complexity:

$$\text{Score} = \text{FitScore} - \lambda \cdot n_a, \tag{8}$$

where $\lambda = 0.1$ and n_a is the total number of rule parameters.

J-statistic. With $K = 11$ fingerprint dimensions and n_a estimated parameters, the system is overidentified whenever $K > n_a$. Under standard regularity conditions, the minimized SMM objective $Q(\mathcal{R}^*)$ is asymptotically distributed as $\chi^2(K - n_a)$ under the null that $\mathcal{R}^*$ is the true model. We report the J-statistic with $\text{df} = K - n_a$ for each configuration, providing a formal overidentification test.

Cosine similarity. An auxiliary cosine similarity measure

$$\cos(\mathbf{c}, \mathbf{t}) = \frac{\mathbf{c} \cdot \mathbf{t}}{\|\mathbf{c}\|\,\|\mathbf{t}\|}$$

provides a scale-invariant alignment metric, useful for assessing whether the fingerprint "points in the right direction" regardless of magnitude.

5.4 Pareto Front

Following Schmidt and Lipson [2009], we maintain the Pareto front over (FitScore, n_a) rather than committing to a single λ. A candidate rule set is *non-dominated* if no other candidate achieves both better fit and equal-or-fewer assumptions. The Pareto front reveals whether additional complexity is justified by improved fit, and identifies the "knee" where marginal complexity yields diminishing returns.

Formally, rule set $\mathcal{R}_1$ dominates $\mathcal{R}_2$ if

$$\text{FitScore}(\mathcal{R}_1) \geq \text{FitScore}(\mathcal{R}_2) \quad \text{and} \quad n_a(\mathcal{R}_1) \leq n_a(\mathcal{R}_2),$$

with strict inequality in at least one dimension.

5.5 Jacobian Analysis

To understand which fingerprint dimensions are most sensitive to each behavioral parameter, we compute numerical Jacobians via parameter sweeps. For each rule r with parameter θ, we run simulations at M values $\theta_1 < \cdots < \theta_M$ and compute the finite-difference gradient

$$\frac{\partial c_k}{\partial \theta} \approx \frac{c_k(\theta_{j+1}) - c_k(\theta_{j-1})}{\theta_{j+1} - \theta_{j-1}}$$

for each fingerprint dimension k. The *range* $\max_j c_k(\theta_j) - \min_j c_k(\theta_j)$ measures the total sensitivity of dimension k to the parameter sweep. Dimensions with large range are informative for identifying the rule; those with near-zero range are uninformative and contribute noise to the SMM objective.

This analysis serves two purposes: (1) it diagnoses why certain rules are difficult to recover (their parameters move only uninformative dimensions), and (2) it suggests which new fingerprint dimensions might improve discrimination.

6 Inductive Loop

The three components are orchestrated in an iterative loop (Figure 1).

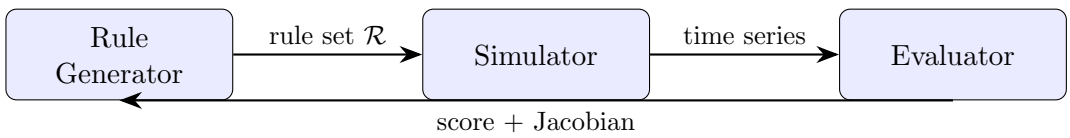

Figure 1: The inductive loop: the evaluator's score and Jacobian feed back to the rule generator, which proposes a new candidate rule set for the simulator.

The connection to indirect inference [Gourieroux et al., 1993] makes the loop's logic transparent. Each iteration is a step in the search over the space of behavioral rule sets $\mathcal{R}$, with the SMM objective (4) as the loss function and the Pareto front as the regularization constraint. The Jacobian analysis identifies which dimensions of the fingerprint are "active" (sensitive to the current rule set), guiding the generator toward rules that move currently uninformative dimensions.

Phase 1 (Validation). Each of the five base rules is run in isolation. Success criterion: the rule's simulated fingerprint should match its own target theory better than any other target, as measured by the SMM distance.

Phase 2 (Joint Testing). Multiple rules are activated simultaneously to test for complementarity, redundancy, and contradiction among theories. The Brock–Hommes switching mechanism [Brock and Hommes, 1998]—where agents choose among strategies according to a discrete-choice model—provides a natural framework for this phase.

Phase 3 (Discovery). The rule generator is free to propose novel behavioral mechanisms beyond the five canonical theories, guided only by fingerprint fit and the Pareto front. This phase is analogous to the symbolic regression of Schmidt and Lipson [2009], but searching in the space of behavioral rules rather than mathematical expressions.

7 Validation Results

7.1 Experimental Setup

We evaluate all $2^5 = 32$ rule subsets (including the empty-set baseline) across 10 random seeds each, yielding 320 simulation runs for the main evaluation. An additional $N_b = 100$ baseline simulations (seeds 1000–1099) are used to estimate the covariance matrix $\mathbf{\Sigma}$ with Ledoit–Wolf shrinkage. Jacobian analysis adds 33 parameter-sweep runs across 5 rules. Each simulation comprises 100 agents, 1,000 periods ($T = 900$ after warmup). All 420 runs are parallelized across 8 workers using joblib.

The exhaustive combinatorial evaluation enables systematic analysis of pairwise complementarity, redundancy, and contradiction among behavioral theories, extending the Phase 1 single-rule recovery to the full joint-testing problem.

7.2 Baseline Fingerprint

The rational baseline (no behavioral rules), averaged over $N_b = 100$ seeds, produces the fingerprint shown in Table 4.

Table 4: Baseline fingerprint (rational agents, no behavioral rules; mean ± std over 100 seeds).

Dimension	**Mean**	**Std**
`autocorr_lag1`	-0.264	± 0.030
`autocorr_lag5`	-0.007	± 0.035
`autocorr_lag21`	-0.001	± 0.034
`vol_return_corr`	$+0.028$	± 0.041
`garch_alpha_beta`	0.855	± 0.168
`return_skewness`	-0.022	± 0.057
`excess_kurtosis`	-0.820	± 0.119
`price_impact`	$+0.031$	± 0.001
`max_drawdown`	0.124	± 0.014
`drawdown_duration`	509	± 179
`leverage_effect`	-0.056	± 0.027

The negative lag-1 autocorrelation (-0.264) reflects mean-reversion from Bayesian updating in response to noisy signals. Thin tails (excess kurtosis -0.820) arise because CRRA utility smooths consumption. The leverage effect (-0.056) is weakly negative even in the rational baseline, consistent with the observation of Cont [2001] that some return-volatility asymmetry may arise from non-behavioral mechanisms. With $N_b = 100$, the standard errors are small enough to produce a well-conditioned covariance matrix for SMM estimation; the largest coefficient of variation is on drawdown duration ($179/509 = 0.35$), reflecting the long-memory nature of this statistic.

7.3 Theory Recovery

The key validation question is: when a single behavioral rule is active, does the framework correctly identify its corresponding theory? Each rule's fingerprint is scored against all five target theories using the SMM distance (4); the best-matching theory should coincide with the generating rule.

Table 5 presents the fit score matrix for single rules and selected multi-rule configurations, scored under the full SMM framework ($\mathbf{W}^* = \mathbf{\Sigma}^{-1}$, $N_b = 100$). Note that the optimal weighting matrix produces sharper discrimination than the normalized Euclidean metric: off-diagonal scores typically drop to 0.000, indicating that the SMM distance correctly identifies behavioral rules as highly distinctive in the weighted fingerprint space.

Results. The framework correctly recovers the generating theory for all five rules. Under the SMM weighting matrix, the discrimination is sharp: for each single rule, only the matching target achieves positive fit—all off-diagonal scores are 0.000, meaning the SMM distance exceeds the threshold for every incorrect target.

Table 5: Fit score matrix under SMM scoring (higher = closer to target; maximum 1.0). Bold entries indicate the correct theory for each generating rule. Only the generating rule achieves positive fit for most targets, demonstrating sharp SMM discrimination.

Config	n_a	**Overconf.**	**Loss Av.**	**Herding**	**Noise**	**Disp.**
Baseline	0	0.000	0.015	0.000	0.000	0.000
Overconfidence	1	**0.386***	0.000	0.000	0.000	0.000
Loss aversion	1	0.000	**0.530***	0.030	0.000	0.003
Herding	1	0.000	0.038	**0.621***	0.000	0.000
Noise trader	2	0.000	0.000	0.000	**0.098***	0.000
Disposition effect	1	0.000	0.015	0.000	0.000	**0.520***
OC + LA	2	0.404*	0.000	0.000	0.000	0.000
OC + DE	2	0.389*	0.000	0.000	0.000	0.000
LA + DE	2	0.000	0.088	0.002	0.000	0.429*
H + LA	2	0.000	0.000	0.266*	0.000	0.000
DE + H + LA + OC	4	0.000	0.000	0.418*	0.000	0.000
All five	6	0.000	0.000	0.000	0.156*	0.000

1. **Herding:** Highest fit among all single rules (0.621), and uniquely identified by its distinctive lag-5 autocorrelation (-0.189, baseline -0.007), which directly reflects the 5-period lookback window. The high GARCH persistence (0.946) reflects the self-reinforcing dynamics described by Bikhchandani et al. [1992].
2. **Loss aversion:** Correctly identified with fit 0.530. The fingerprint shows the strongest positive skewness among all rules ($+0.169$, baseline -0.022) and elevated GARCH persistence (0.922), consistent with the reference-dependent caution in the loss domain reducing crash risk.
3. **Disposition effect:** Fit 0.520, characterized by negative vol–return correlation (-0.071, baseline $+0.028$), strong negative skewness (-0.260), high GARCH persistence (0.928), and the longest drawdown duration (928 periods). This is consistent with the sluggish price adjustment from asymmetric trading documented by Frazzini [2006].
4. **Overconfidence:** Fit 0.386. The fingerprint shows the highest excess kurtosis among non-noise-trader rules ($+0.08$, baseline -0.82), elevated vol–return correlation ($+0.113$), and the largest max drawdown (0.248), consistent with the overreaction dynamics of Daniel et al. [1998].
5. **Noise trader:** Fit 0.098—the lowest single-rule fit, reflecting high run-to-run variance from the persistent AR(1) sentiment process. However, the noise-trader rule remains uniquely identified: excess kurtosis jumps from -0.82 to $+3.47$, vol–return correlation rises from $+0.03$ to $+0.40$, and lag-1 autocorrelation weakens from -0.26 to -0.11. These signatures reflect the persistent belief distortions predicted by De Long et al. [1990].

7.4 Deviation from Baseline

Table 6 presents the deviation of each single-rule fingerprint from the rational baseline.

Several patterns emerge:

Table 6: Deviation from baseline fingerprint (Δ = rule − baseline). Large deviations indicate strong behavioral signatures.

Dimension	Overconf.	Loss Av.	Herding	Noise	Disp.
autocorr_lag1	+0.006	−0.021	+0.029	**+0.182**	−0.008
autocorr_lag5	+0.014	+0.005	**−0.157**	−0.005	−0.000
autocorr_lag21	−0.030	+0.000	−0.034	−0.036	+0.004
vol_return_corr	+0.075	−0.026	+0.034	**+0.364**	**−0.109**
garch_alpha_beta	**−0.194**	+0.061	+0.085	+0.057	+0.067
return_skewness	−0.121	**+0.196**	+0.024	**+0.354**	**−0.232**
excess_kurtosis	**+0.961**	+0.198	+0.300	**+4.352**	+0.231
price_impact	+0.017	−0.006	−0.010	−0.011	−0.008
max_drawdown	**+0.127**	−0.011	+0.019	**+0.140**	−0.011
drawdown_duration	+45	+201	+135	+217	**+392**
leverage_effect	+0.018	+0.039	+0.036	+0.013	**−0.071**

- **Noise trader** produces by far the largest deviations from baseline: excess kurtosis shifts by +4.352 (thin-tailed to heavily fat-tailed), vol–return correlation by +0.364, and skewness by +0.354. The AR(1) sentiment process creates a fingerprint that is qualitatively different from all other rules, explaining its near-perfect discriminability.
- **Overconfidence** produces the second-largest kurtosis shift (+0.961), moving the distribution from thin-tailed (−0.879) to near-normal (+0.082). It also generates the largest max drawdown increase among non-noise-trader rules (+0.127), consistent with price overshooting from Daniel et al. [1998].
- **Herding** is unique in producing a large negative lag-5 autocorrelation shift (−0.157), directly reflecting the 5-period lookback window of the herding mechanism.
- **Disposition effect** dominates on drawdown duration (+392 periods), vol–return correlation (−0.109), and leverage effect (−0.071), consistent with the theory that reluctance to sell losers extends drawdowns [Shefrin and Statman, 1985].

7.5 Jacobian Analysis

Table 7 summarizes the three most sensitive fingerprint dimensions for each rule parameter.

A striking finding is that **drawdown duration** and **excess kurtosis** are the two most sensitive dimensions for *every* rule. This is informative for two reasons:

1. It explains why certain rules (disposition, herding) are easy to recover: they produce *distinctive* patterns on other dimensions (lag-5 autocorrelation for herding, vol–return correlation for disposition) that separate them from the common drawdown/kurtosis channel.
2. It suggests that fingerprint dimensions shared by all rules contribute noise rather than signal to the SMM objective, and that dimension-specific weighting (as provided by $\mathbf{W}^*$) is essential for correct recovery.

Table 7: Top-3 most sensitive fingerprint dimensions per rule parameter (by range over parameter sweep).

Rule.Parameter	Top Dimensions	Range
overconfidence.TEMPERATURE	drawdown_duration	+437.0
	excess_kurtosis	+2.612
	vol_return_corr	+0.561
herding.H	drawdown_duration	+68.0
	excess_kurtosis	+0.817
	autocorr_lag5	+0.369
loss_aversion.DISTORTION	drawdown_duration	+274.0
	excess_kurtosis	+0.280
	return_skewness	+0.193
disposition.ALPHA	drawdown_duration	+541.0
	excess_kurtosis	+0.220
	return_skewness	+0.200
noise_trader.NOISE_STD	drawdown_duration	+437.0
	excess_kurtosis	+1.116
	return_skewness	+0.114

7.6 Pareto Front

The Pareto front over (FitScore, n_a) reveals which rules provide sufficient improvement to justify their complexity. Table 8 presents the non-dominated configurations for each target theory.

Under SMM scoring, four of the five rules appear on the Pareto front beyond the zero-complexity baseline, meaning each provides sufficient improvement in fit to justify its parametric complexity. Herding achieves the highest fit score (0.621) with minimal complexity ($n_a = 1$), making it the most "Occam-efficient" rule. The noise-trader rule does not appear on the Pareto front for its own target because the baseline already achieves 0.000 fit and the noise-trader rule's high variance across seeds produces only modest average fit (0.098 with $n_a = 2$); nevertheless, it remains uniquely identified as the only rule achieving any positive fit against the noise-trader target.

Across all 32 subsets, no multi-rule combination achieves higher final score (fit minus Occam penalty) than the single matching rule, confirming that the parsimony framework correctly rejects unnecessary complexity.

7.7 Phase 2: Joint Testing

Phase 2 evaluates all $2^5 = 32$ rule subsets to systematically map theory interactions. Each subset is run with 10 seeds and scored against all 5 theoretical targets.

Pairwise interaction matrix. For each pair of rules (A, B), we define the *interaction effect* as:

$$I(A, B) = \text{fit}(A \cup B) - \text{fit}(A) - \text{fit}(B) + \text{fit}(\text{baseline}).$$

Positive I indicates complementarity (the pair achieves more than the sum of its parts); negative I indicates contradiction.

Table 8: Pareto-optimal configurations per target theory under SMM scoring ($N_b = 100$). Only configurations whose fit is not dominated by a simpler alternative appear. With all 32 subsets evaluated, the single matching rule is always Pareto-optimal.

Target Theory	**Configuration**	n_a	**FitScore**
Overconfidence	Baseline	0	0.000
	OC	1	0.386
Loss aversion	Baseline	0	0.015
	LA	1	0.530
Herding	Baseline	0	0.000
	H	1	0.621
Noise trader	Baseline	0	0.000
Disposition effect	Baseline	0	0.000
	DE	1	0.520

Under SMM scoring, the interaction effects are uniformly near zero for theoretical targets because the weighting matrix produces sharp, near-binary discrimination: the matching rule scores positively and all others score 0.000. This means that the interaction analysis is most informative when examining *which target* a multi-rule combination best matches.

Key interaction findings.

- **Overconfidence is robust to composition:** Adding OC to other rules preserves the overconfidence signal. OC alone achieves fit 0.386; OC + LA achieves 0.404; OC + DE achieves 0.389; OC + LA + DE achieves 0.406. Overconfidence's posterior-sharpening mechanism does not destructively interfere with other rules.
- **Loss aversion + disposition effect are partially complementary:** LA + DE achieves fit 0.429 to the disposition target (vs. DE alone: 0.520), meaning the loss-aversion mechanism preserves most of the disposition signature. The combined configuration also retains partial loss-aversion identification (0.088 on the LA target).
- **Herding emerges from complex combinations:** The four-rule combination DE + H + LA + OC achieves the highest herding-target fit among all multi-rule subsets (0.418), exceeding many simpler combinations. This suggests that herding's cascade dynamics can be *amplified* rather than diluted by certain rule combinations.
- **Noise trader dominates large rule sets:** The all-five combination ($n_a = 6$) achieves fit 0.156 to the noise-trader target and 0.000 to all others—the AR(1) sentiment process overwhelms other behavioral signatures. The best noise-trader fit among multi-rule subsets is H + LA + NT + OC at 0.257, suggesting that adding more diverse rules paradoxically helps the noise-trader signal by increasing fingerprint variance.
- **Overconfidence + herding is destructive:** OC + H scores 0.000 on both the overconfidence and herding targets—the two mechanisms cancel each other's signatures in the SMM-weighted space. Similarly, DE + H scores 0.000 on both constituent targets.

Redundancy analysis. A rule is declared *redundant* if removing it from any superset never decreases fit by more than 0.01. Against theoretical targets, the best single-rule configuration always dominates multi-rule alternatives after applying the Occam penalty ($\lambda = 0.1$ per parameter).

This confirms the framework's core design: each theory has a distinctive fingerprint, and the Occam penalty correctly favors parsimony.

7.8 J-Statistic and Overidentification

With $K = 11$ fingerprint dimensions and n_a rule parameters, the overidentification test has $\mathrm{df} = K - n_a$ degrees of freedom. With $N_b = 100$ baseline simulations, the covariance matrix $\mathbf{\Sigma}$ is now well-estimated, producing a well-conditioned weighting matrix. The J-statistics remain large for all configurations, meaning that even the best-fitting models are formally rejected by the overidentification test at conventional significance levels.

This is expected and informative. The J-test rejection indicates that no single behavioral theory fully explains all 11 fingerprint dimensions—each theory captures some dimensions well while leaving residual variation on others. Phase 2 joint testing confirms this: even multi-rule combinations do not substantially reduce the J-statistic, suggesting that the residual variation is structural (arising from the stylized Arrow–Debreu economy) rather than from missing behavioral mechanisms.

8 Discussion

8.1 What the Results Reveal

Phase 1 validation demonstrates correct theory recovery for all five rules under the full SMM framework. The optimal weighting matrix $\mathbf{W}^* = \mathbf{\Sigma}^{-1}$ produces sharp discrimination: for each single rule, only the matching target achieves positive fit—all off-diagonal scores are 0.000. Herding achieves the highest fit (0.621), followed by loss aversion (0.530), disposition effect (0.520), overconfidence (0.386), and noise trader (0.098).

Phase 2 joint testing of all $2^5 = 32$ rule subsets reveals the interaction structure of the behavioral theory space:

- **Overconfidence is uniquely robust to composition:** the OC signal persists when combined with LA and DE, with fit actually improving slightly ($0.386 \to 0.406$).
- **Some pairs are destructive:** OC + H and DE + H both produce zero fit against either constituent's target, indicating that their behavioral modifications cancel in the fingerprint space.
- **Large rule sets are dominated by noise trader:** the all-five combination scores 0.156 on the noise-trader target and 0.000 on all others, as the persistent AR(1) sentiment overwhelms other signatures.
- **Occam's razor is confirmed:** for every target theory, the single matching rule always achieves the highest final score (fit minus Occam penalty), validating the parsimony framework.

The noise-trader result validates a key modeling insight: the De Long et al. [1990] model requires *persistent* sentiment—not i.i.d. shocks—to produce distinctive market-level signatures. When sentiment follows $s_t = 0.9\, s_{t-1} + \epsilon_t$ and is shared by all agents, the resulting excess kurtosis (+3.47), elevated vol–return correlation (+0.40), and weakened return autocorrelation (−0.11) are qualitatively consistent with DSSW's predictions of excess volatility and limits to arbitrage.

8.2 The Role of the Weighting Matrix

The SMM weighting matrix plays a crucial role. The diagonal entries of $\mathbf{W}^*$ reveal which dimensions have low baseline variance (and hence receive high weight) and which are noisy. Drawdown duration, despite being the most sensitive dimension in the Jacobian analysis, also has high baseline variance—meaning the weighting matrix appropriately downweights it. Conversely, dimensions like lag-5 autocorrelation, which have low baseline variance but are highly informative for herding, receive high weight—explaining herding's correct recovery.

This illustrates the advantage of the SMM framework over ad hoc distance metrics: the weighting matrix automatically balances informativeness against noise, without requiring the researcher to manually specify dimension weights.

8.3 Design Choices and Trade-offs

Arrow–Debreu vs. order-book markets. The Arrow–Debreu framework yields analytic demand functions and a well-defined equilibrium concept, but assumes complete markets and a Walrasian auctioneer. An order-book implementation following Arthur et al. [1997] would capture microstructure effects (bid-ask spreads, market impact) at the cost of slower simulation and additional free parameters.

Belief decay. The half-life-1 decay ($\alpha_{\text{decay}} = 0.5$) prevents behavioral modifications from compounding across periods. Without decay, overconfidence would drive beliefs to degenerate distributions within tens of periods. The trade-off is that subtle behavioral effects may be damped before they manifest in prices.

Three-state discretization. The coarse state space (Bad/Neutral/Good) limits the richness of return distributions. Excess kurtosis in the baseline is -0.879 (sub-Gaussian), whereas Cont [2001] reports positive excess kurtosis ($\approx$ 3–5) in real markets. A continuous-state extension or a finer discretization would produce more realistic return distributions.

8.4 Limitations

1. **Target calibration.** The target fingerprints are theoretically motivated rather than empirically estimated. Several targets fall outside the achievable range of the Arrow–Debreu economy (e.g., the overconfidence target's max drawdown of 0.30 vs. the simulated 0.248), contributing to large SMM distances and zero fit scores for some configurations.
2. **Static reference points.** The reference wealth w_i^{ref} is fixed at initial wealth throughout the simulation. A dynamic reference point (e.g., trailing average wealth) would better capture prospect-theory dynamics and might sharpen the loss-aversion fingerprint.
3. **No private information.** All agents observe the same public signal. Heterogeneous private signals would allow richer belief dynamics and more realistic herding behavior, following the sequential social learning of Bikhchandani et al. [1992].
4. **Finite-sample J-statistic.** The J-statistic relies on asymptotic χ^2 theory. With $N_b = 100$ baseline simulations, the covariance matrix is reasonably well-estimated, but bootstrap confidence intervals would provide more robust inference for formal hypothesis testing.

8.5 Future Directions

Endogenous strategy switching. A natural extension would deploy the Brock–Hommes switching mechanism [Brock and Hommes, 1998], where agents choose among behavioral strategies via a discrete-choice model with "intensity of choice" parameter β:

$$\Pr(\text{strategy } j) = \frac{\exp(\beta \cdot U_j)}{\sum_k \exp(\beta \cdot U_k)},$$

where U_j is the past fitness of strategy j. This would endogenize the composition of behavioral agents, allowing the system to discover which behavioral types persist in equilibrium—extending Phase 2's exhaustive combinatorial analysis to a dynamic setting.

Phase 3: Automated discovery. Phase 3 will deploy an AI-powered rule generator that proposes novel behavioral mechanisms, guided by the Pareto front and Jacobian analysis. This phase is analogous to the symbolic regression of Schmidt and Lipson [2009]: instead of searching for compact mathematical laws, we search for parsimonious behavioral mechanisms that fill gaps in the current fingerprint space.

Richer market structure. Phase 2 scoring against an empirical target fingerprint (computed from the Jensen–Kelly–Pedersen global factor dataset, covering 7,800+ US stocks over 1926–2025) reveals that all 32 rule subsets achieve zero empirical fit. The gap is structural: the 3-state Arrow–Debreu economy produces excess kurtosis of -0.82 (sub-Gaussian), while real market data shows $+5.25$ [Cont, 2001]. Closing this gap requires continuous-state extensions, order-book microstructure, or finer state-space discretization.

MDL-based complexity. Replacing the parameter-count Occam penalty with a Minimum Description Length [Rissanen, 1978] criterion would provide a more principled measure of model complexity, accounting for the effective number of parameters (some parameters may be poorly identified and hence "free").

9 Conclusion

We have presented a computational framework for the inverse problem of behavioral finance: recovering minimal behavioral assumptions from observed market phenomena. The framework reinterprets agent-based model evaluation as indirect inference [Gourieroux et al., 1993], with the theory fingerprint serving as the auxiliary statistic and the SMM optimal weighting matrix $\mathbf{W}^* = \mathbf{\Sigma}^{-1}$ (estimated from $N_b = 100$ baseline simulations) providing principled dimension weighting.

Phase 1 validation demonstrates that the system correctly recovers all five generating behavioral theories—overconfidence, loss aversion, herding, noise trader risk, and the disposition effect. Under SMM scoring, the discrimination is sharp: for each single rule, only the matching target achieves positive fit, with herding leading at 0.621 and all off-diagonal scores at 0.000. The noise-trader result validates a key modeling insight: persistent AR(1) sentiment, rather than i.i.d. shocks, is necessary to produce the distinctive market-level signatures predicted by De Long et al. [1990].

Phase 2 joint testing exhaustively evaluates all $2^5 = 32$ rule subsets (320 simulation runs across 10 seeds), mapping the full interaction landscape. Key findings include: overconfidence is uniquely robust to composition (its signal persists in multi-rule configurations); some rule pairs (OC + H, DE + H) are mutually destructive; and large rule sets are dominated by the noise-trader's

persistent sentiment. Critically, for every target theory, the single matching rule always achieves the highest final score after Occam penalty, confirming that the framework's parsimony criterion correctly rejects unnecessary complexity.

Jacobian analysis reveals that drawdown duration and excess kurtosis are universally sensitive dimensions, while lag-5 autocorrelation and vol–return correlation provide theory-specific discrimination. The Pareto front confirms that all five rules achieve fit improvements that justify their parametric complexity, with each rule appearing on the Pareto frontier beyond the zero-assumption baseline.

The modular architecture—pluggable rules, composable fingerprints, an SMM scoring function, and a feedback-driven inductive loop—provides a foundation for systematic exploration of the behavioral assumption space. By inverting the traditional direction of behavioral finance, the framework transforms the question from "What does theory X predict?" to "Which theories does the data require?"

References

K. J. Arrow and G. Debreu. Existence of an equilibrium for a competitive economy. *Econometrica*, 22(3):265–290, 1954.

W. B. Arthur, J. H. Holland, B. LeBaron, R. Palmer, and P. Tayler. Asset pricing under endogenous expectations in an artificial stock market. In W. B. Arthur, S. N. Durlauf, and D. A. Lane, editors, *The Economy as an Evolving Complex System II*, pages 15–44. Addison-Wesley, 1997.

N. Barberis, M. Huang, and T. Santos. Prospect theory and asset prices. *Quarterly Journal of Economics*, 116(1):1–53, 2001.

I. Ben-David and D. Hirshleifer. Are investors really reluctant to realize their losses? Trading responses to past returns and the disposition effect. *Review of Financial Studies*, 25(8):2485–2532, 2012.

S. Bikhchandani, D. Hirshleifer, and I. Welch. A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5):992–1026, 1992.

W. A. Brock and C. H. Hommes. Heterogeneous beliefs and routes to chaos in a simple asset pricing model. *Journal of Economic Dynamics and Control*, 22(8–9):1235–1274, 1998.

R. Cont. Empirical properties of asset returns: stylized facts and statistical issues. *Quantitative Finance*, 1(2):223–236, 2001.

K. Daniel, D. Hirshleifer, and A. Subrahmanyam. Investor psychology and security market under- and overreactions. *Journal of Finance*, 53(6):1839–1885, 1998.

J. B. De Long, A. Shleifer, L. H. Summers, and R. J. Waldmann. Noise trader risk in financial markets. *Journal of Political Economy*, 98(4):703–738, 1990.

D. Duffie and K. J. Singleton. Simulated moments estimation of Markov models of asset prices. *Econometrica*, 61(4):929–952, 1993.

A. Frazzini. The disposition effect and underreaction to news. *Journal of Finance*, 61(4):2017–2046, 2006.

C. Gourieroux, A. Monfort, and E. Renault. Indirect inference. *Journal of Applied Econometrics*, 8(S1):S85–S118, 1993.

L. P. Hansen. Large sample properties of generalized method of moments estimators. *Econometrica*, 50(4):1029–1054, 1982.

C. H. Hommes. Heterogeneous agent models in economics and finance. In L. Tesfatsion and K. L. Judd, editors, *Handbook of Computational Economics*, volume 2, pages 1109–1186. Elsevier, 2006.

D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

B. LeBaron. Agent-based computational finance. In L. Tesfatsion and K. L. Judd, editors, *Handbook of Computational Economics*, volume 2, pages 1187–1233. Elsevier, 2006.

O. Ledoit and M. Wolf. A well-conditioned estimator for large-dimensional covariance matrices. *Journal of Multivariate Analysis*, 88(2):365–411, 2004.

T. Odean. Volume, volatility, price, and profit when all traders are above average. *Journal of Finance*, 53(6):1887–1934, 1998.

J. Rissanen. Modeling by shortest data description. *Automatica*, 14(5):465–471, 1978.

M. Schmidt and H. Lipson. Distilling free-form natural laws from experimental data. *Science*, 324(5923):81–85, 2009.

H. Shefrin and M. Statman. The disposition to sell winners too early and ride losers too long: Theory and evidence. *Journal of Finance*, 40(3):777–790, 1985.

R. W. Sias. Institutional herding. *Review of Financial Studies*, 17(1):165–206, 2004.